# Class 12.03.2025

1) basic models in 1D in population dynamics
2) Bifurcation theory (tangent bifurcation).

# Autonomous systems of one dimension

$$(A)\begin{cases} \dot{x} = f(x) \\ x(0) = x_0 \end{cases}, \quad x \in \mathbb{R} \ (\text{or on } I \subset \mathbb{R} \text{ open})$$

$f: \mathbb{R} \longmapsto \mathbb{R}$ regular

↳ vector field

(A) is autonomous as $f$ does not depend on $t$.

phase space: { value of $x$ }




The vector field $f(x)$ is generating a flow that transports the initial datum $x_0$ into its evolved at time $t$. So, we use the notation

$$\boxed{x(t) = \underline{\Phi}^t(x_0)}, \quad t \in \mathbb{R}$$

* If $t \in \mathbb{R}$, $\exists$ a map

$$\phi^t : M \longrightarrow M$$ diffeomorfism (same regularity of the vector field)

having the following properties

(1) $\forall\, t, s \in \mathbb{R}$, $\phi^{t+s} = \phi^t \circ \phi^s = \phi^s \circ \phi^t$

i.e. (commutativity)

$$\phi^{t+s}(x) = \phi^t(\phi^s(x)) = \phi^s(\phi^t(x))$$

(2) $\forall\, t \in \mathbb{R}$, $(\phi^t)^{-1} = \phi^{-t}$ ($\exists$ inverse)

i.e.

$$\phi^{-t}(\phi^t(x)) = x$$

(3) $\phi^0 = Id$ i.e. $\phi^0(x) = x$

## Rmk

If $t \in \mathbb{R}_+$, then $\Phi^t$ is a semi-flow, in the sense that (2) does not hold true.

## Linear equation

$$\begin{cases} \dot{x} = \lambda x, \\ x(t_0) = x_0 \end{cases} \qquad \lambda \in \mathbb{R}$$

* $x_0 = 0 \implies x(t) \equiv 0, \ \forall \lambda \in \mathbb{R}$
* $x_0 \neq 0 \implies \begin{cases} x(t) \equiv x_0, & \lambda = 0 \\ x(t) = x_0 e^{\lambda(t-t_0)}, & \lambda \neq 0 \end{cases}$

$\Downarrow t_0 = 0$

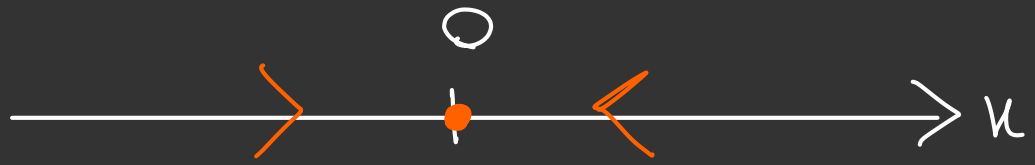
$\Phi^t = e^{\lambda t}$

$$x(t) = x_0 e^{\lambda(t-t_0)}, \quad x(t) \xrightarrow{t\to+\infty} +\infty, \quad \lambda > 0$$

$$x(t) \xrightarrow{t\to+\infty} 0, \quad \lambda < 0$$

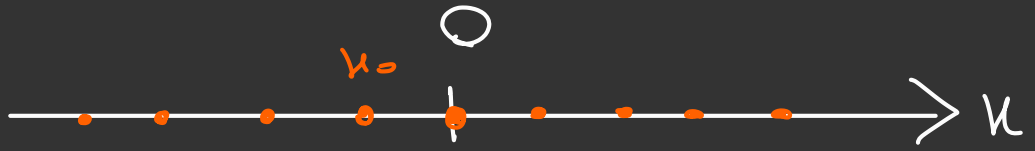
$0$ ↑ equilibrium

$\lambda < 0$



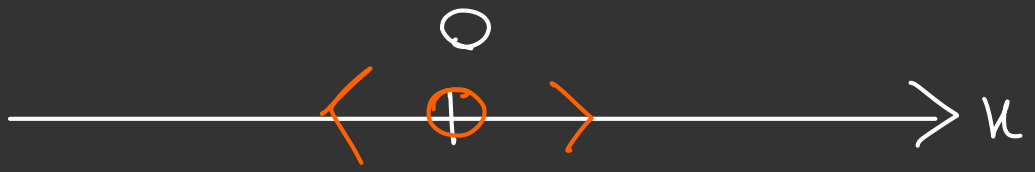

attractive equi

$\lambda = 0$



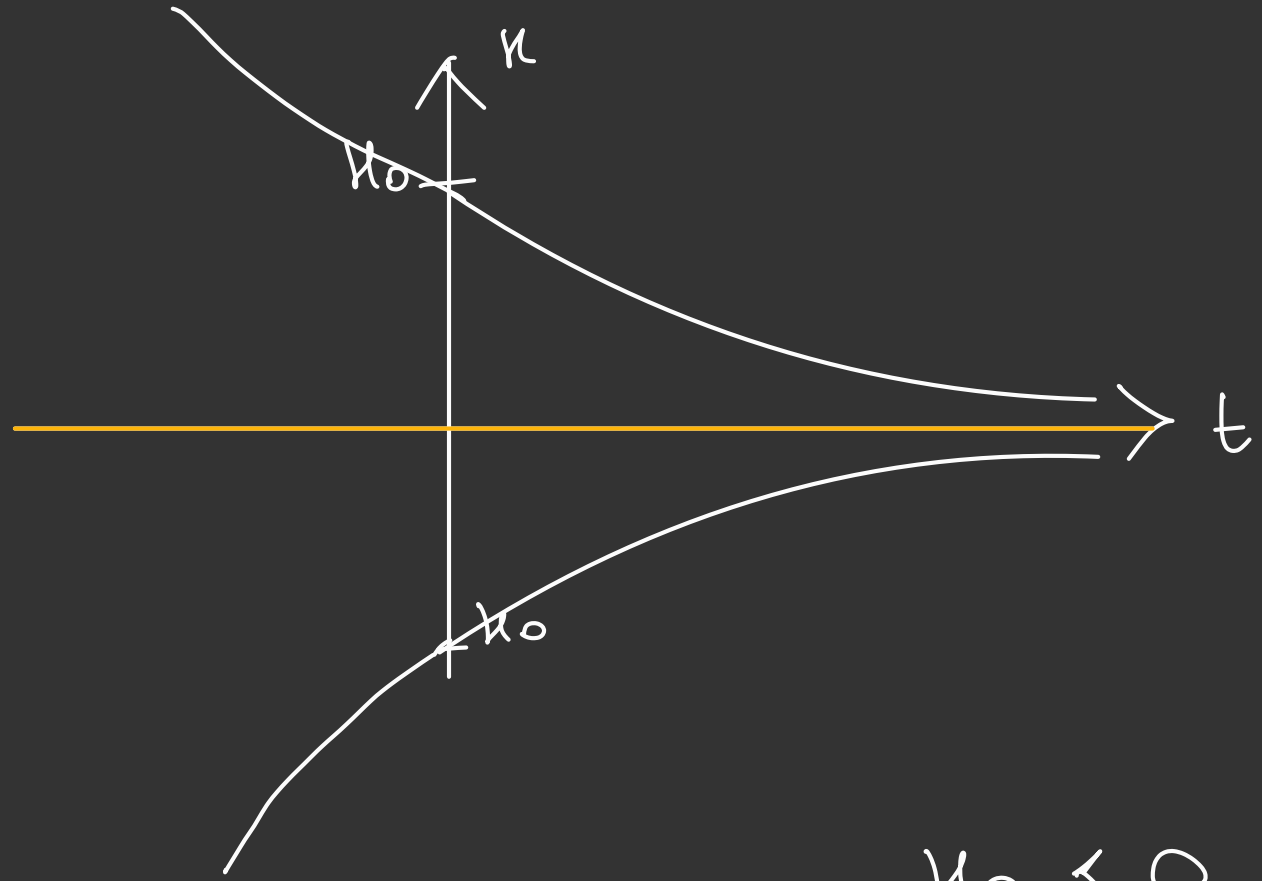

only stationary sol ($\forall x_0 \in \mathbb{R}$ is a sol)

$\lambda > 0$




repulsive equi (or attractive in the past).

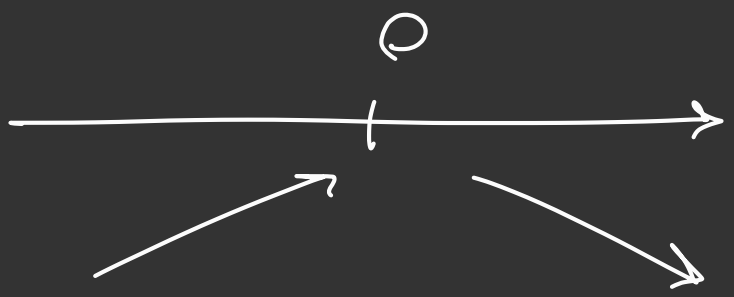
$\lambda = 0$ bifurcation parameter: it changes the nature of the equilibrium.



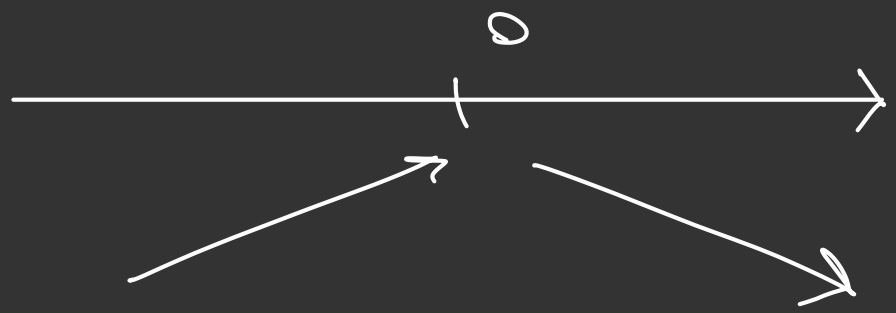

$\dot{x} = \lambda x$

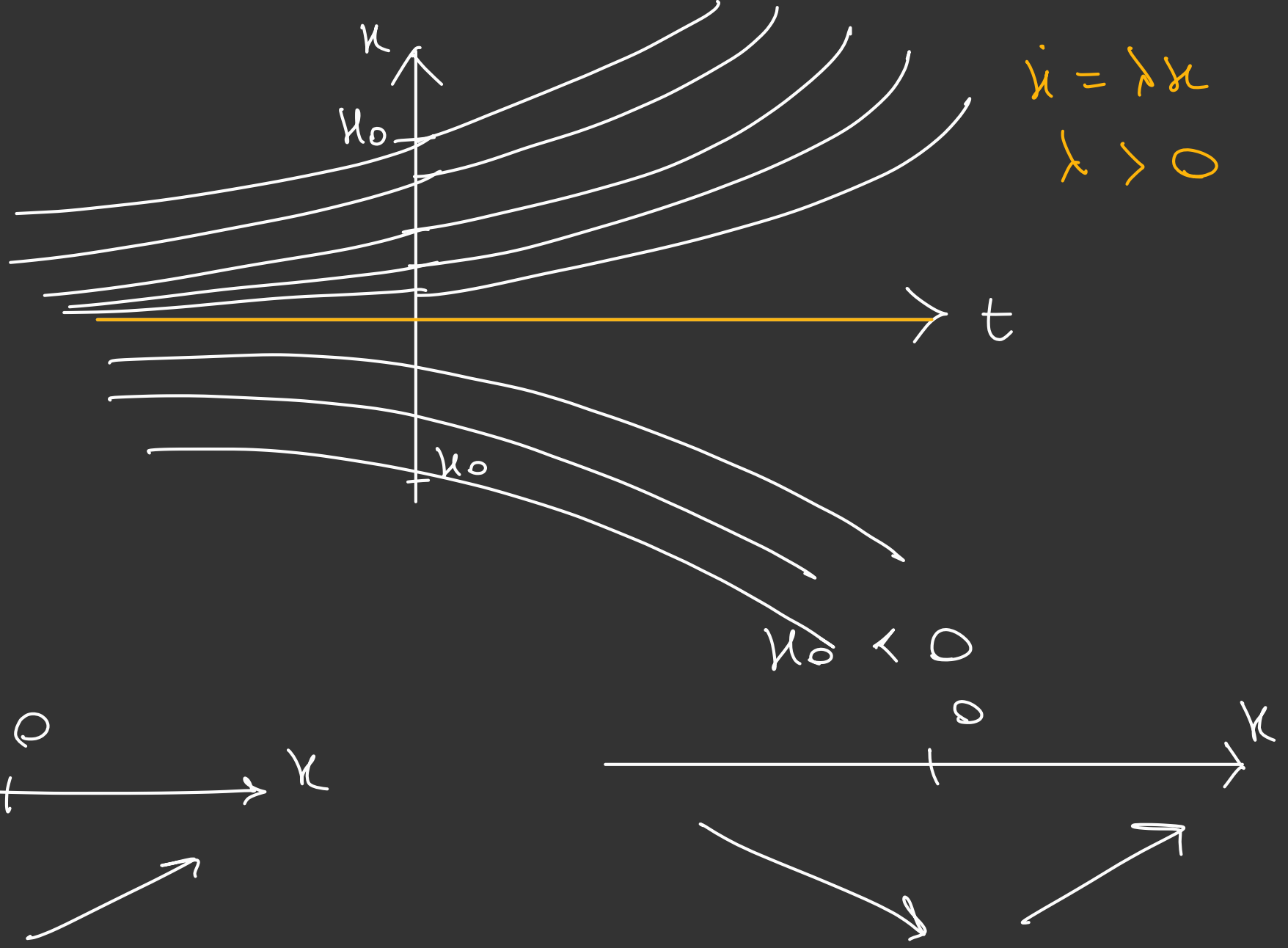
$\lambda < 0$

$x_0 > 0$




$x_0 < 0$

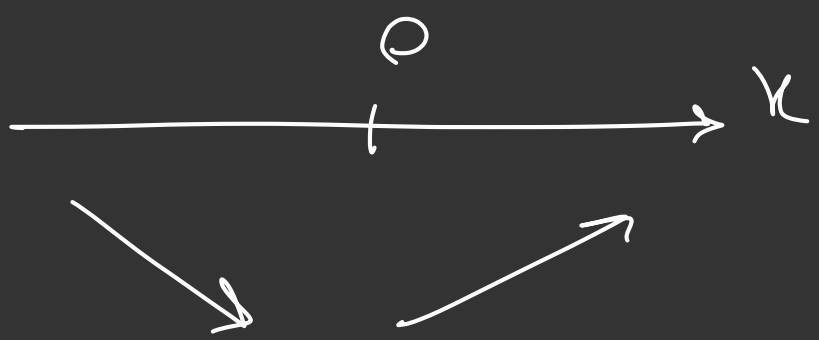




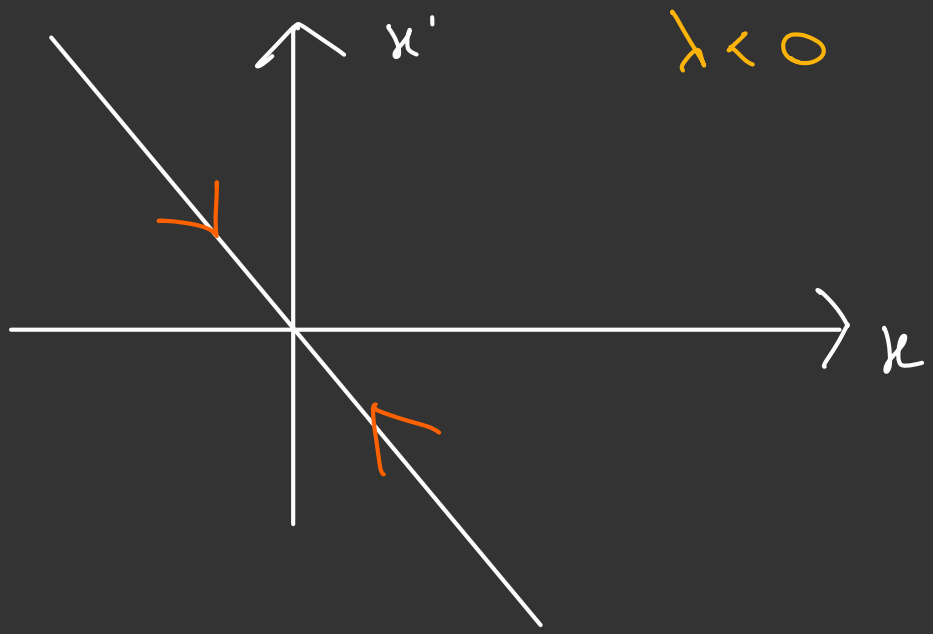

$\dot{x} = \lambda x$

$\lambda > 0$

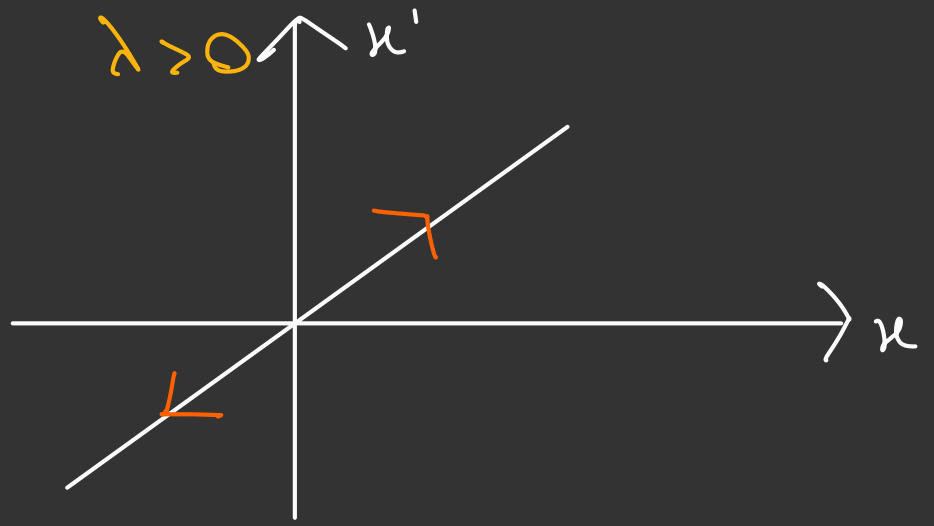
$x_0 > 0$




$x_0 < 0$

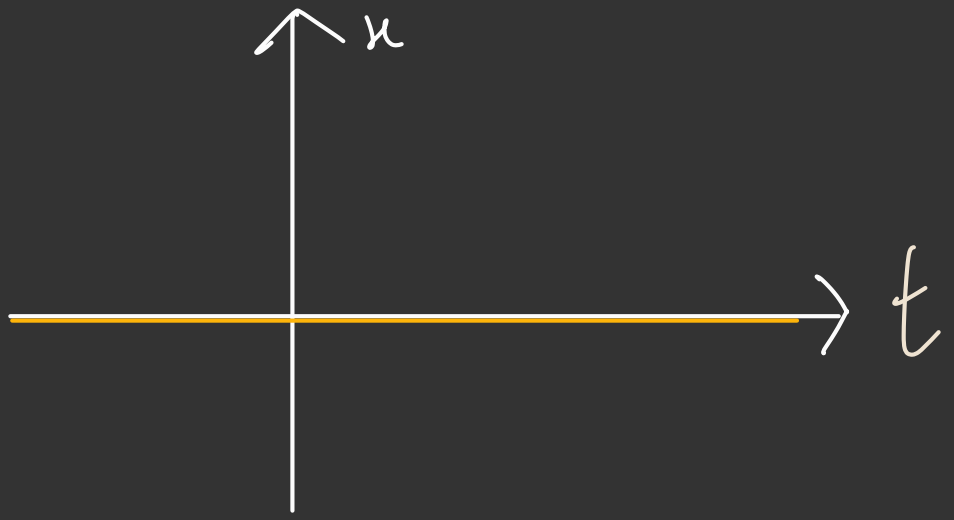



$\lambda < 0$




$\lambda > 0$

$\lambda = 0$



$\lambda < 0$ ⇝ exponential decay of radio-active substances

$\lambda > 0$ ⇝ population growth (growth rate > death rate) where there's an illimited availability of ressource.

1 [∃ steady states] ↝ stationary solution $D \subset \mathbb{R}$

We consider $\dot{x} = f(x)$, (*) $f \in C^2(\mathbb{R})$

A stationary solution $\bar{x}$ of (*) is ~~iff~~ iff $f(\bar{x}) = 0$

$\bar{x} \in \mathbb{R}$

(Then, $\bar{x} \in \mathbb{R}$ is a stationary sol (*) iff $f(\bar{x}) = 0$).

Prop 2 [Stability steady states]

Let $\bar{x}$ be a stat. sol of (*)

Then, it holds

(1) $f'(\bar{x}) > 0 \Rightarrow \bar{x}$ is ~~a repulsive equilibrium~~ unstable.

(2) $f'(\bar{x}) < 0 \Rightarrow \bar{x}$ is ~~attractive equilibrium~~ asymptotically stable.

(Q): $f'(\bar{x}) = 0$ ? ↝ bifu analysis.

DEF: * $\bar{x}$ is stable if

$\forall \varepsilon > 0$ : $B_\varepsilon(\bar{x}) \subset D$ s.t. $\exists \delta$ s.t. $\phi^t(x) \in B_\varepsilon(\bar{x})$ $\forall t \geq 0, x \in B_\delta(\bar{x})$

$\varepsilon \in (0, \varepsilon]$

* $\bar{x}$ is unstable if it is not stable.

* $\bar{x}$ is asymptotically stable if

$\bar{x}$ is stable and if $\lim_{t \to +\infty} \phi^t(x) = \bar{x}$, $\forall x \in B_\delta(\bar{x})$

# Bifurcation theory

$$\dot{x} = f_\mu(x) \quad , x \in D \subset \mathbb{R}$$

$f$ depends regularly on $\mu$

NOT: $f_\mu(x) = f(\mu, x)$

The bifurcation analysis (BA) gives an answer to the question "What does happen when $f'(\bar{x}) = 0$.

Assume $\exists\ \mu_c, \bar{x}$ s.t. $f(\mu_c, \bar{x}) = 0 = f'(\mu_c, \bar{x})$

↝ 3 scenarios

1) Tangent bifurcation.
2) Bifurcation with change of stability.
3) Pitchfork bifurcation.

1) It is so called because when $\mu = \mu_c$ $f_{\mu_c}(\bar{x})$ is tangent at the axes $x$ (the orizontal axes).

Ex: $\dot{x} = (x - \hat{x})^2 - \mu =: f_\mu(x)$

$f_\mu(x) = 0 \iff (x - \hat{x})^2 = \mu$

↝ (i) $\mu = 0 \Rightarrow \bar{x} = \hat{x}$

(ii) $\mu < 0 \Rightarrow \nexists\ \bar{x}$ s.t. $f_\mu(\bar{x}) = 0$

(iii) $\mu > 0 \Rightarrow \bar{x}_\pm = \hat{x} \pm \sqrt{\mu}$

$f'_\mu(x) = 2(x - \hat{x}) = 0 \iff x = \hat{x} \Rightarrow \mu_c = 0$

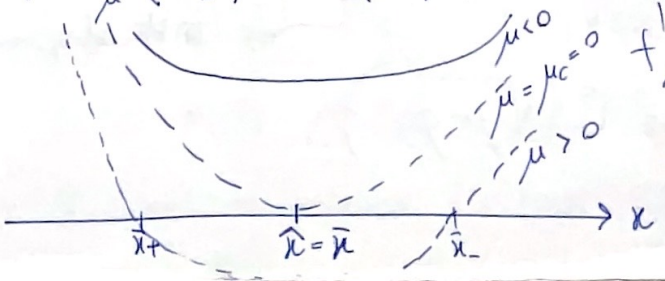
$f''_\mu(x) = 2 > 0$

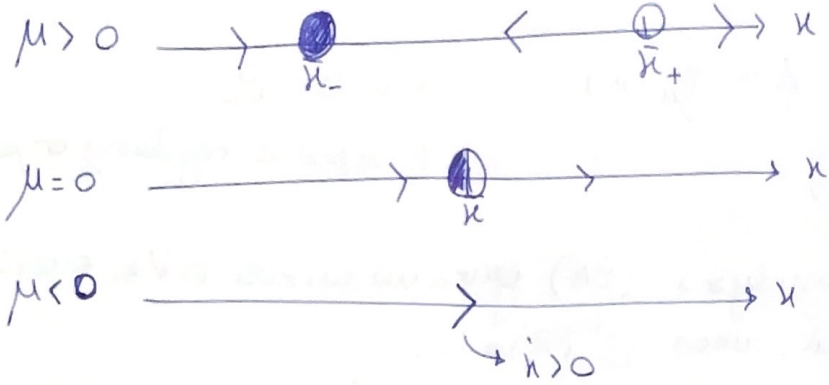
(i) $f'_{\mu_c}(\bar{x}) = 0$ ⚠

(ii) $\nexists$ steady state

(iii) $f'_\mu(\bar{x}_\pm) = 2 \cdot (\pm\sqrt{\mu}) \rightsquigarrow f'_\mu(\bar{x}_+) > 0$

$f'_\mu(\bar{x}_-) < 0$






Therefore, we can conclude that we have a tangent bifurcation if $f_{\mu_c}(\bar{x}) = f'_{\mu_c}(\bar{x}) = 0$

$f''_{\mu_c}(\bar{x}) \neq 0$ and $\partial_\mu f_\mu(\bar{x})\big|_{\mu=\mu_c} \neq 0$

we will understand this condition in the statement and proof below.

Prop 1

We consider del DS $\dot{x} = f_\mu(x)$, $x \in D \subset \mathbb{R}$, $\mu \in \mathbb{R}$

We assume $\exists\, \mu_c, \bar{x}$ s.t. ($f$ regular enough wrt $x, \mu$)

$$f_{\mu_c}(\bar{x}) = f'_{\mu_c}(\bar{x}) = 0$$

and $f''_{\mu_c}(\bar{x}) > 0$, $\partial_\mu f_\mu(\bar{x})\big|_{\mu=\mu_c} < 0$ (in general $f''_{\mu_c}(\bar{x})\,\partial_\mu f_\mu(\bar{x})\big|_{\mu_c=\mu} < 0$)

Then,

for $\mu \in B_{\mu_c}$ $\exists$ tangent bifurcation s.t.

(i) $\nexists$ steady state for $\mu < \mu_c$

(ii) $\exists\, \bar{x}_1(\mu), \bar{x}_2(\mu) \in \mathbb{R}$ steady states for $\mu > \mu_c$ s.t.

$$\bar{x}_1(\mu) < \bar{x} < \bar{x}_2(\mu)$$

$\bar{x}_1$: stable, $\bar{x}_2$: unstable

with $|\bar{x}_i(\mu) - \bar{x}| = \mathcal{O}\left(\sqrt{\mu - \mu_c}\right)$.